# Deep Reinforcement Learning from Human Preferences

2017.07

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# 1. Introduction

- → Previous problems
  - → Not clear how to construct a suitable reward function.
    - → Ex) To train a robot to clean a table or scramble an egg.
  - → Not applicable to behaviors that are difficult for humans to demonstrate.
    - → Controlling a robot with freedom and non-human morphology

# 2. Methods

- → Providing human feedbacks.
  - → Requiring hundreds of hours of experience.
    - → Focusing on decreasing the amount of feedback required.

# 2. Methods

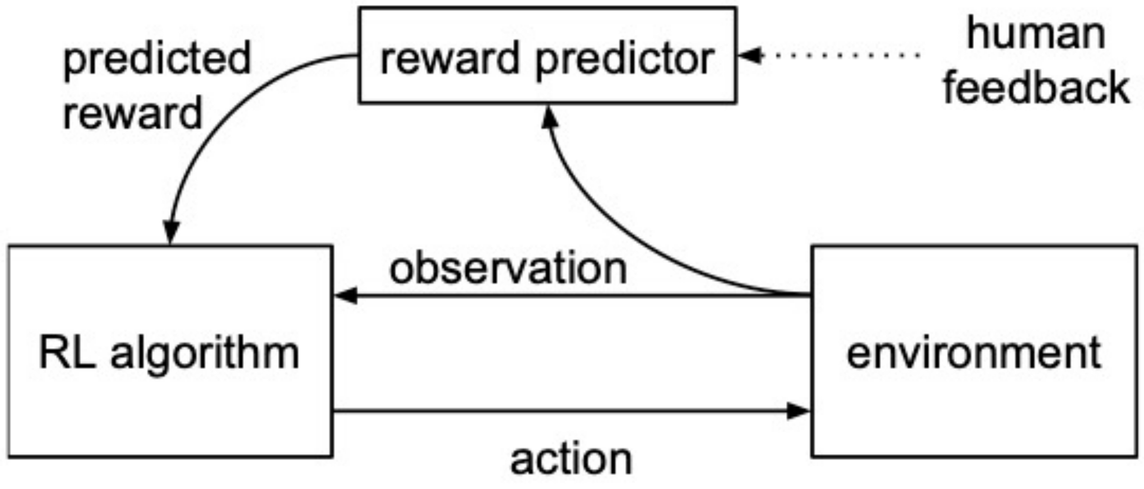
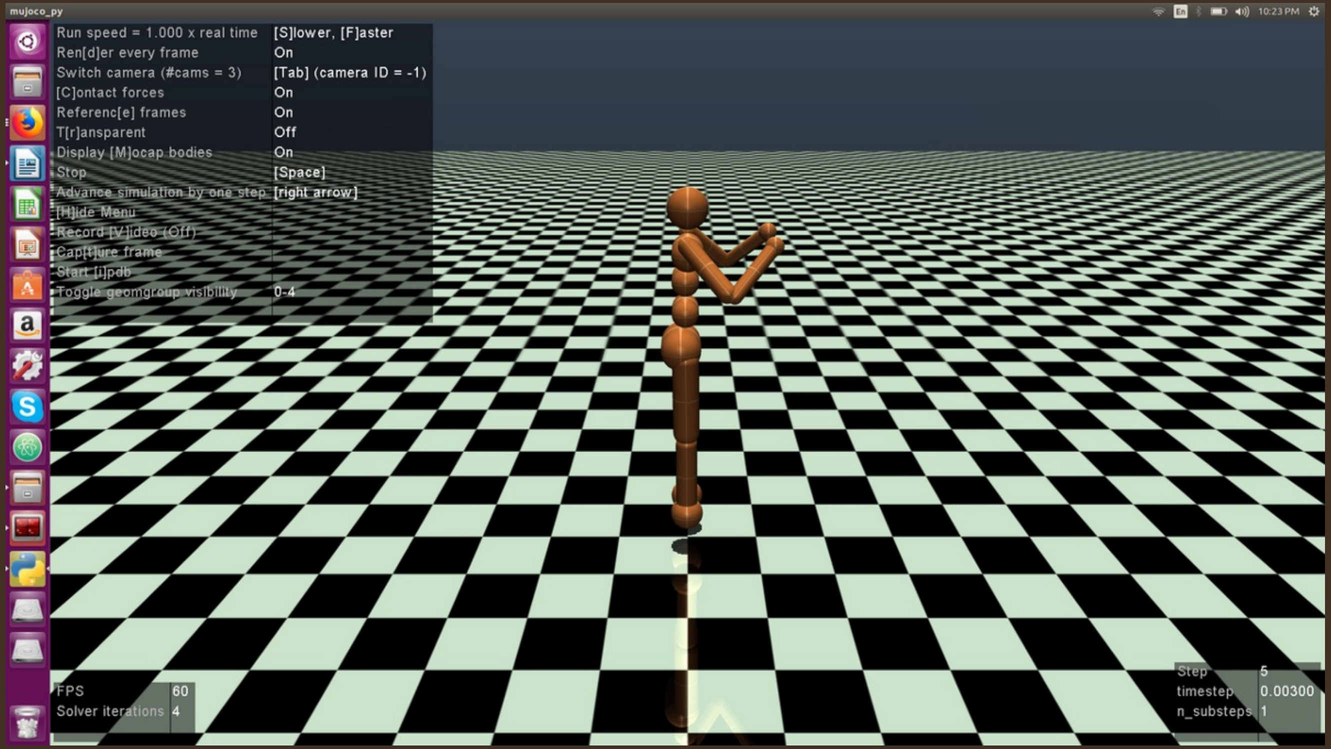



Figure 1: Schematic illustration of our approach: the reward predictor is trained asynchronously from comparisons of trajectory segments, and the agent maximizes predicted reward.

1. enables us to solve tasks for which we can only recognize the desired behavior, but not necessarily demonstrate it.
2. allows agents to be taught by non-expert users.
3. scales to large problems, and
4. is economical with user feedback.

# 2. Methods

Arcade Learning




MuJoCo

# 2. Methods

Trajectory segments : a sequence of observation states and actions states

$$\sigma = ((o_0, a_0), (o_1, a_1), \ldots, (o_{k-1}, a_{k-1})) \in (\mathcal{O} \times \mathcal{A})^k$$

Quantitative evaluation from human's preferences

$$((o_0^1, a_0^1), \ldots, (o_{k-1}^1, a_{k-1}^1)) \succ ((o_0^2, a_0^2), \ldots, (o_{k-1}^2, a_{k-1}^2))$$

From a reward function (r) -> Maximize the discounted sum of rewards.

$$r(o_0^1, a_0^1) + \cdots + r(o_{k-1}^1, a_{k-1}^1) > r(o_0^2, a_0^2) + \cdots + r(o_{k-1}^2, a_{k-1}^2).$$

# 2. Methods

## How to train?

1. The policy $\pi$ interacts with the environment to produce a set of trajectories $\{\tau^1, \ldots, \tau^i\}$. The parameters of $\pi$ are updated by a traditional reinforcement learning algorithm, in order to maximize the sum of the predicted rewards $r_t = \hat{r}(o_t, a_t)$.
2. We select pairs of segments $(\sigma^1, \sigma^2)$ from the trajectories $\{\tau^1, \ldots, \tau^i\}$ produced in step 1, and send them to a human for comparison.
3. The parameters of the mapping $\hat{r}$ are optimized via supervised learning to fit the comparisons collected from the human so far.

Running asynchronously !!

(1) -> (2) : generating trajectories
(2) -> (3) : human-feedback
(3) -> (4) : r-hat updating

# 2. Methods

Training the reward function?

Assuming human's judgments preferring a segment on the value of the latent reward summation

$$\hat{P}\left[\sigma^1 \succ \sigma^2\right] = \frac{\exp \sum \hat{r}\left(o_t^1, a_t^1\right)}{\exp \sum \hat{r}(o_t^1, a_t^1) + \exp \sum \hat{r}(o_t^2, a_t^2)}.$$

Minimize the cross-entropy loss between predictions and human labels

$$\operatorname{loss}(\hat{r}) = -\sum_{(\sigma^1,\sigma^2,\mu)\in\mathcal{D}} \mu(1) \log \hat{P}\left[\sigma^1 \succ \sigma^2\right] + \mu(2) \log \hat{P}\left[\sigma^2 \succ \sigma^1\right].$$

# 3. Experiments & Results

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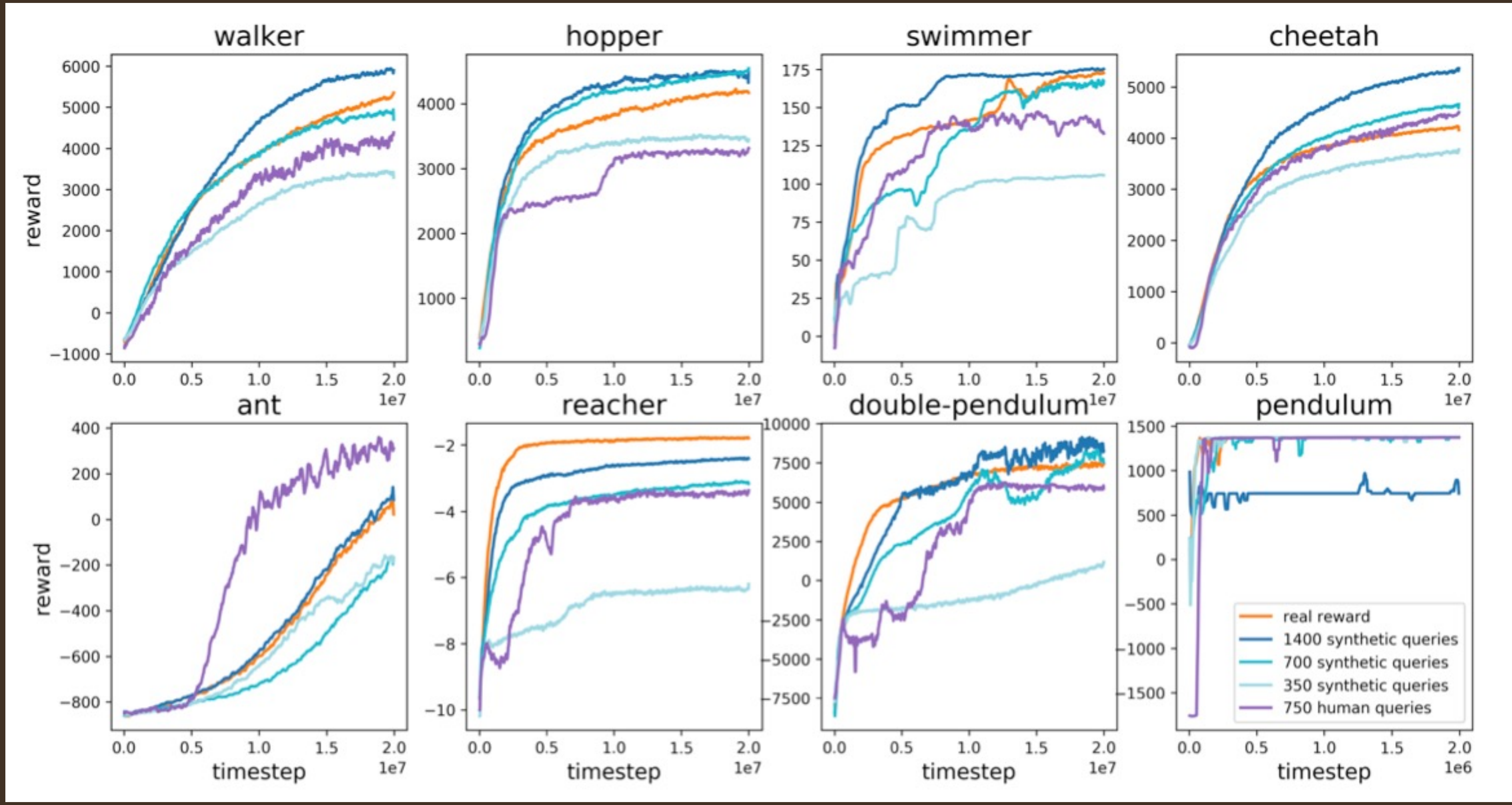


- Training agent with 700 queries to a human rater.
- Comparing to learning from 350, 700 or 1400 synthetic queries.

# 3. Experiments & Results

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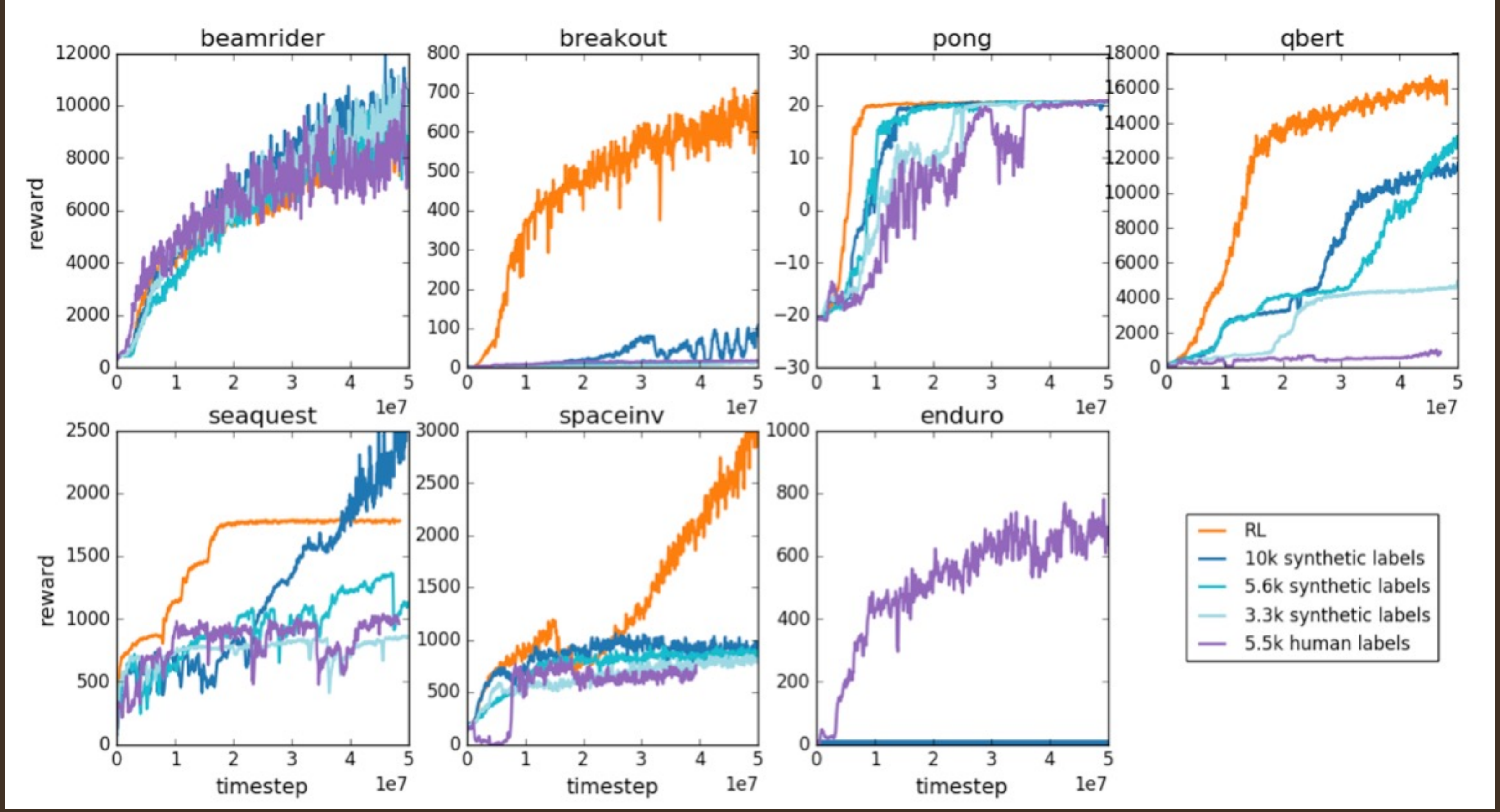


- → Training agent with 5500 queries to a human rater.
- → Comparing to learning from 350, 700 or 1400 synthetic queries.

# Discussion & Conclusion

- → Meaningfully train deep RL agents with human preferences.
- → Improvements of on further sample-complexity and reduces the cost of non-expert feedback.

Thank you!